

Supporting Information

Table of Content

1. Sequences of oligonucleotides used in the study
2. Detailed experimental procedures
3. Design of *M. smegmatis*-specific sensor and its performance with total *M. smegmatis* RNA
4. Analysis of the inhibited peroxidase-like deoxyribozyme cleavage in gel
5. Performance of MTC- and *M. smegmatis*-specific sensors using synthetic analytes
6. Electrophoretic analysis of total RNA isolated from *M. bovis* BCG and *M. smegmatis*
7. Limits of detection for MTC and *M. smegmatis*-specific sensors using total bacterial RNA
8. Specificity of MTC- and *M. smegmatis*-specific sensors with synthetic analytes
9. References

1. Sequences of oligonucleotides used in the study

Table S1. Names and sequences of the oligodeoxyribonucleotides used in this study.

Name	Sequence
Mtb	5'- CAC GGG ATG CAT GTC TTG TGG TGG AAA GCG CTT TAG CGG TGT GGG ATG AG
M.smeg	5'- CCT GCT GGT CGC ATG GCC TGG TAG GGG AAA GCT TTT GCG GTG TGG GAT GG
Dz _a (MTC)	5'- CTC ATC CCA CAC CGC TAA AGC GCTT A CAA CGA <u>GAA CCC AAC C</u>
Dz _b (MTC)	5'- GT TGC TCA TGG A <u>GG CTA GCT TCC ACC ACA AGA CAT GCA TCC CGT G</u>
Dz _a (<i>M.smeg</i>)	5'- CCA TCC CAC ACC GCA AAA GCT TTCC A CAA CGA <u>GAACCC AAC C</u>
Dz _b (<i>M.smeg</i>)	5'- GT TGC TCA TGG A <u>GG CTA GCT CCT ACC AGG CCA TGC GAC CAG CAG G</u>
IPDz	5'- GGG TAG GGC GGG TTG GGT TC rG rU CC ATG AGC AACTCG CCC
PDz	5'- GGG TAG GGC GGG TTG GGT TCG

Nucleotides constituting the catalytic core of 10-23 deoxyribozyme are shown in italics. Nucleotides of split deoxyribozyme strands, Dz_a and Dz_b, that are complimentary to IPDz are underlined. Ribonucleotides are in blue.

2. Detailed experimental procedures.

RNA isolation. Total RNA was isolated from either *M. smegmatis* (strain MC2 155) or *M. bovis* (strain BCG) as previously published.^[1] The concentration of total RNA was calculated by measuring its absorption at 260 nm using a Nanodrop spectrophotometer and taking into account that 1 OD₂₆₀ corresponds to 40 µg/mL. The stock concentrations were 660 ng/µL and 1400 ng/µL for BCG and *M. smegmatis* RNA, respectively.

Gel electrophoresis. Total bacterial RNA isolated either from *M. smegmatis* (strain MC2 155) or from *M. bovis* (strain BCG) (1 µg in 5 µL) was mixed with 5 µL of a loading buffer×2 containing 2 mM EDTA, 0.01% bromophenol blue, 7 M urea, 13% Ficoll (w/v), 90 mM Tris-borate, pH 8.3. To serve as a size marker, ssRNA Ladder (New England BioLabs Inc, MA) (1 µg) was used. The samples were heated at 65 °C for 5 minutes, chilled on ice and loaded onto a 1% agarose gel. The gel was run at 100 V for 1 h, stained with GelRed (Biotium, CA), and the bands were visualized using a U:Genius gel documentation system (Syngene, MD).

Colorimetric assay. IPDz (1 µM), Dz_a (1 µM), Dz_b (30 nM) were mixed with either synthetic analyte or total bacterial RNA in the buffer containing 50 mM HEPES, pH 7.4, 50 mM MgCl₂, 20 mM KCl, 120 mM NaCl, 0.03% Triton X-100, 1% DMSO (reaction buffer) and incubated at 50 °C for 1 h. A typical reaction volume was 120 µL. In case of total bacterial RNA, the samples were heated at 95 °C for 5 min

prior to incubation to ensure proper hybridization of the deoxyribozyme strands to 16S rRNA. After incubation, the mixtures were cooled down to room temperature for 15 min to allow PDz to form a G-quadruplex structure. Then, hemin and DAB were added to the mixtures to the final concentration of 200 nM and 1 mM, respectively. Finally, H₂O₂ was added to the final concentration of 1 mM, and the mixture were incubated at room temperature for 30 min, then the absorbance of the final samples at 500 nm was measured using a Lambda 35 UV/Vis Spectrometer (Perkin Elmer, Inc., MA). The data was processed using Microsoft Excel 2010.

TLC plate-based visualization. To visualize an analyte-dependent color change, 3 μ L of the final samples were spotted on a silica gel TLC plate (EMD Chemicals Inc., Germany) and let dry on air. The plate was then scanned using a CanoScan LiDE210 scanner (Canon, NY), and the image was analyzed using a Gel-Pro Analyzer software. To calculate the relative intensity (RI) of the spots, the intensity of a sample was divided by the intensity of 100 nM Dz, which was used as a positive control for hemin-mediated DAB oxidation. The brightness and contrast of the image was adjusted using Corel PHOTO-Paint X6 software (Corel, Canada) to ensure proper visualization in a printed form.

3. Design of *M. smegmatis*-specific sensor and its performance with total *M. smegmatis* RNA.

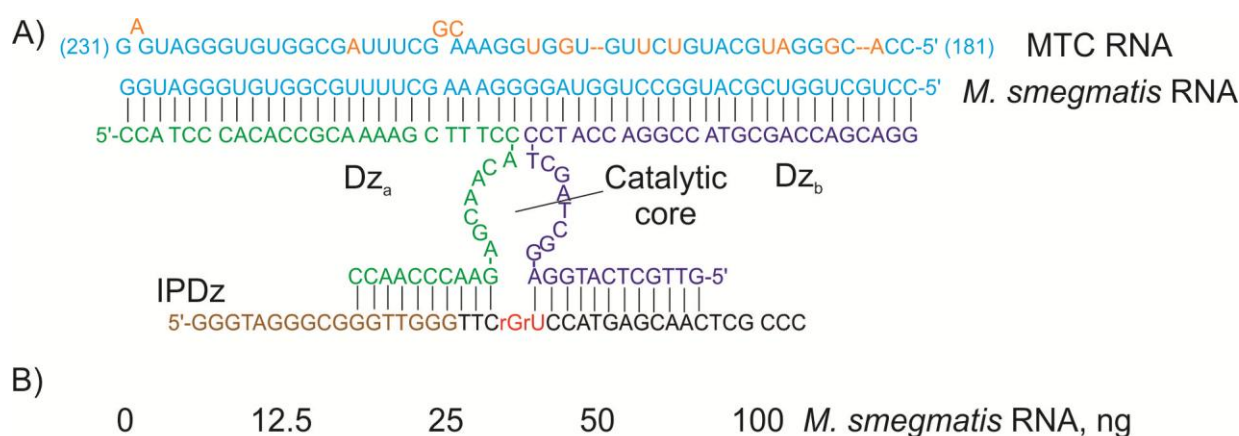


Figure S1. Deoxyribozyme cascade for *M. smegmatis* detection. A) Design of *M. smegmatis*-specific deoxyribozyme sensor. Sequences of *M. smegmatis* and *M. tuberculosis* 16S rRNA differ by several nucleotide substitutions (shown in orange). Analyte-binding arms of the split deoxyribozyme strands are complementary to *M. smegmatis* sequence. IPDz substrate is shown bound to the deoxyribozyme strands. Ribonucleotide inserts of IPDz are in red. B) TLC plate-based visualization of the signal for *M. smegmatis*-specific sensor in the presence of different amount of total RNA from *M. smegmatis*.

The design of the binary deoxyribozyme (Dz) probes was accomplished according to the previously published procedure by Mokany et al.^[2] Briefly, the inhibited peroxidase-like deoxyribozyme (IPDz)-binding arms and the split catalytic core of Dz_a and Dz_b was the same for two different sensors, and only the analyte-binding arms of Dz_a and Dz_b were designed complementary to their specific analytes (Figures 2a and S1a). The targeted fragment of MTC analyte contained eight single-nucleotide and one dinucleotide substitutions, as well as two single-nucleotide and one dinucleotide insertions/deletions compared to the sequence of *M. smegmatis* analyte. These differences in the analyte sequences allow the sensors to differentiate between the two mycobacteria.

For the current assay, a segment of 16S rRNA that has significant nucleotide differences has been chosen, which ensured perfect differentiation of the two mycobacterial species. However, earlier we have demonstrated that structural adjustments of split probes may allow differentiation of analytes that differ

by a single nucleotide.^[3,4] Therefore, the suggested Dz cascade is potentially applicable for the analysis of single nucleotide polymorphisms and point mutations, which opens a way toward simple and inexpensive sensors for diagnosing human genetic disorders, as well as drug-resistant bacteria.

4. Analysis of the inhibited peroxidase-like deoxyribozyme (IPDz) cleavage in gel

Analysis of the cleavage of the IPDz by binary Dz was performed by polyacrylamide gel electrophoresis (PAGE). A representative data is shown in Figure S1. It was observed that binary Dz cleaved IPDz into two fragments (21 nt and 18 nt) in the presence of the cognate analyte (Lanes 3-7). The intensity of the band corresponding to the 21-nt PDz increased with the increase in analyte concentration. The cleavage was negligible if no analyte was added (Lane 2). It proved that the catalytic core of the deoxyribozyme is re-formed only by the formation of complete tetrapartite complex shown in Figure S2A.

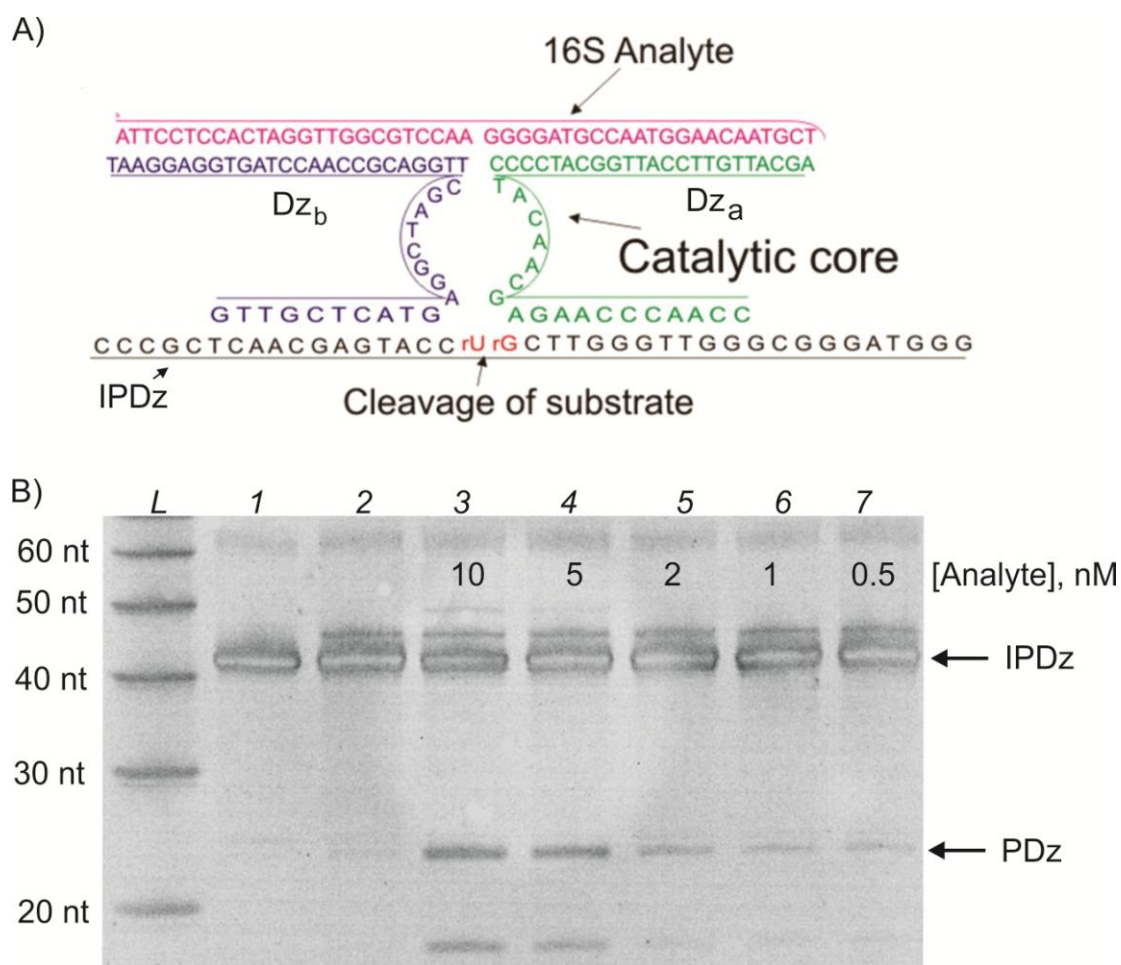


Figure S2. Analysis of the products of IPDZ cleavage by binary 10-23 probe. A) Binary 10-23 probe in complex with IPDz substrate and the cognate analyte. B) Analysis of the cleavage products in 15% urea polyacrylamide gel. IPDz (1 μ M), Dz_a (1 μ M) and Dz_b (30 nM) were incubated in the presence of different concentrations of the 16S analyte for 1h at 55 °C, (lanes 3-7). Lane L- 10-nt DNA ladder (IDT, Coralville); Lane 1, 1 μ M IPDz; Lane 2: IPDz (1 μ M), Dz_a (1 μ M) and Dz_b (30 nM) (no analyte, negative control). The positions of IDPz and the cleavage product PDz are indicated by the arrows on the right.

5. Performance of MTC- and *M. smegmatis*-specific sensors using synthetic analytes

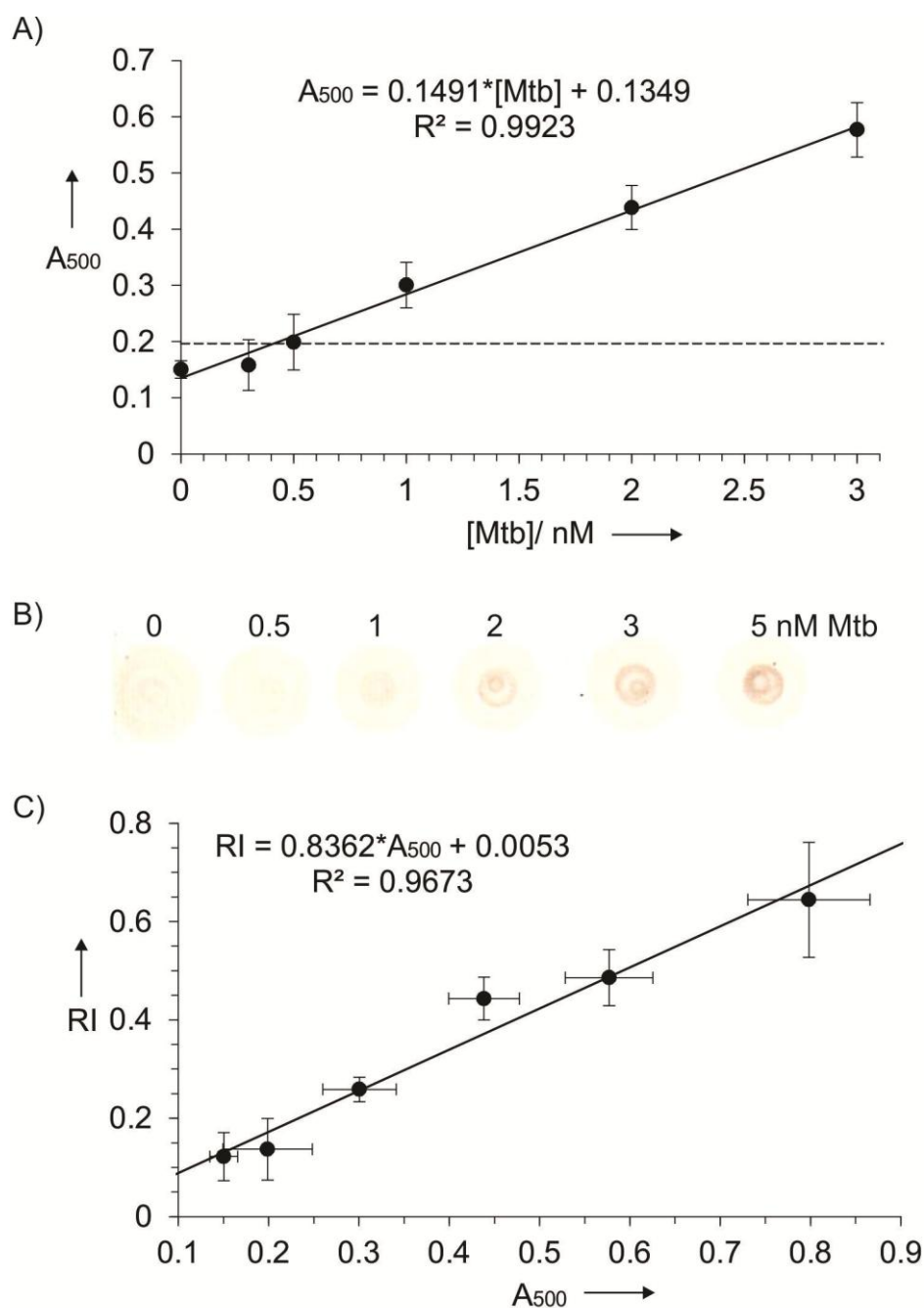


Figure S3. Detection of synthetic Mtb analyte using MTC-specific sensor. A) Absorbance of the sensor at 500 nm in the presence of different Mtb analyte concentrations. The dashed line represents absorbance values correspondent to the limit of detection.³ The data is the average from three independent experiments, the error bars are given with standard deviations. B) TLC plate-based visualization of the signal for MTC-specific sensor in the presence of different Mtb concentrations. C) Correlation between absorbance (A_{500}) values presented at panel (A) and relative intensity (RI) values of the spots from panel (B) in the presence of different concentration of Mtb analyte. RI was calculated as the intensity of the spot at definite analyte concentration divided by the intensity of the control spot, which contained 1 μM PDz instead a mixture of IPDz, Dz_a, Dz_b and analyte. The data is the average from three independent experiments, the error bars are given with standard deviations.

First, we tested the performance of the designed sensors using synthetic oligodeoxyribonucleotides mimicking the targeted fragments of bacterial 16S rRNAs (Table S1). The presence of the analyte induced the cleavage of IPDz and release of an active peroxide-like Dz into solution. The addition of hemin, DAB and H₂O₂ resulted in the color-change of the analyte-containing samples, which was monitored by measuring the absorbance of the samples at 500 nm. It was observed that the absorbance increased linearly with the increase in the analyte concentration in the range of 0-5 nM, with the limit of detection (LOD) calculated to be 0.4 nM and 0.6 nM for MTC- and *M. smegmatis*-specific sensors, respectively (Figures S3aA and S4A). This LOD is ~ 100 times lower than that obtained with PDZ-based sensor without the DZ cascade.^[4] Since an instrument-free detection format is preferable for the point-of-care diagnostic assay, we next explored a TLC plate-based format for the color-change visualization. For this purpose, 3 μ L of the samples were spotted on a TLC plate (Figures S3B and S4B). The intensity of the spots correlated well with the absorbance values (Figure S3C). However, only 1 nM analyte concentration allowed distinguishing the presence of the analyte at the LOD from the background with the naked eye.

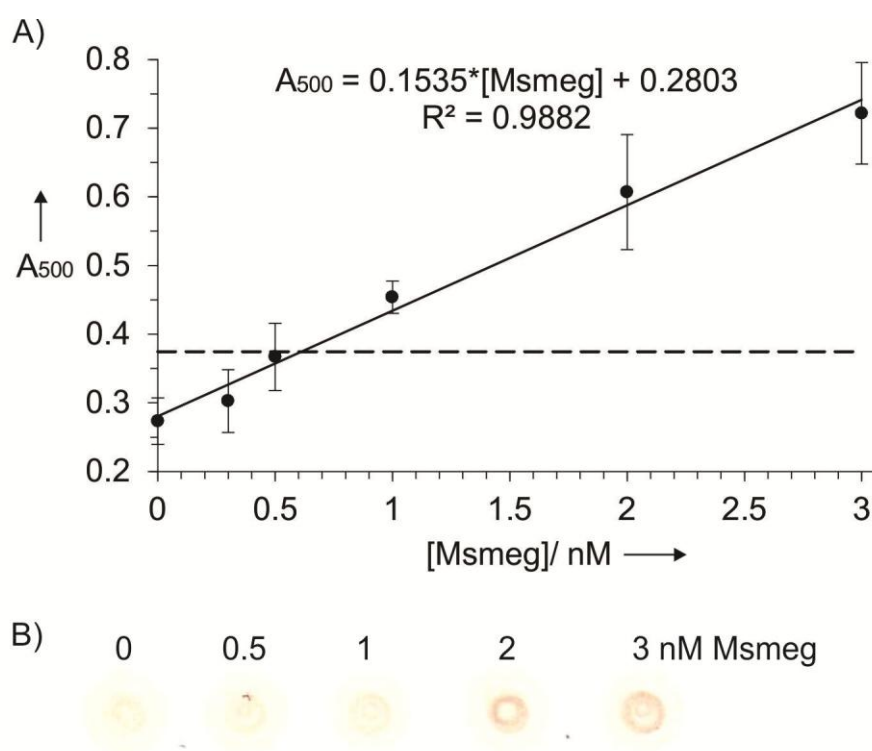


Figure S4. Detection of synthetic Msmeg analyte using *M. smegmatis*-specific sensor. A) Absorbance of the sensor at 500 nm in the presence of different Msmeg concentrations. The dashed line represents absorbance values correspondent to the limit of detection. The data is the average from three independent experiments, the error bars are given with standard deviations. B) TLC plate-based visualization of the signal for *M. smegmatis*-specific sensor in the presence of different Msmeg concentrations.

6. Electrophoretic analysis of total RNA isolated from *M. bovis* BCG and *M. smegmatis*

In order to check if total RNA preparations from BCG and *M. smegmatis* contain similar content of 16S rRNA, the RNA samples of the same concentration were analyzed by electrophoresis in 1% agarose gel. The gel was stained with ethidium bromide. It can be seen that the intensity of 16S rRNA bands (1542 nt) for both bacterial species is close to each other.

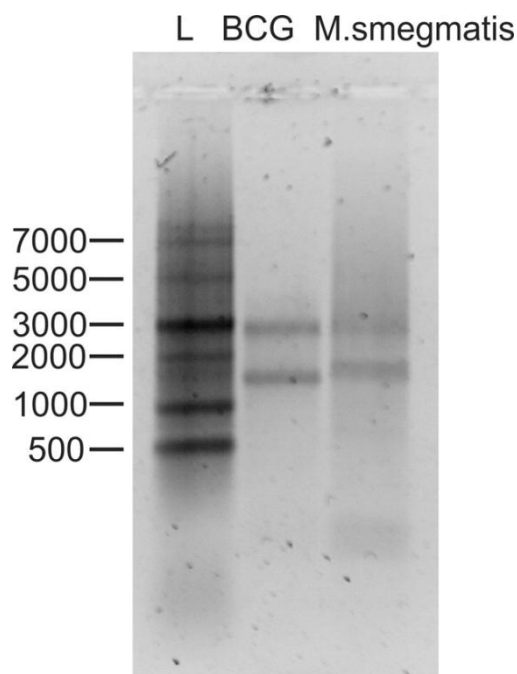


Figure S5. 1% agarose gel-electrophoresis of total RNA preparations from BCG strain of *M. bovis* and *M. smegmatis*. L – ssRNA ladder (the sizes of the fragments in nucleotides are shown on the left).

7. Limits of detection for *M. smegmatis*-specific sensors using total bacterial RNA

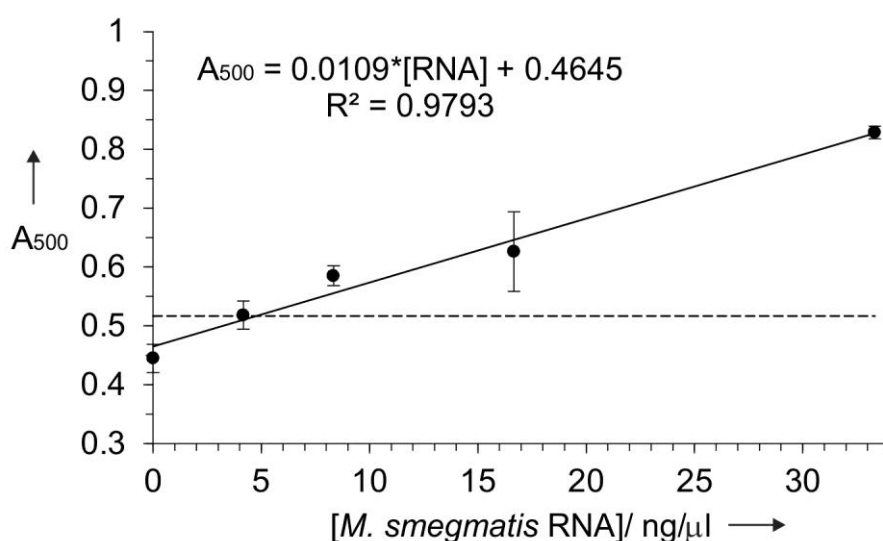


Figure S6. Detection limits for *M. smegmatis*-specific sensors in case of total RNA isolated from *M. smegmatis*. The dashed line represents absorbance values correspondent to the limit of detection. The data is the average from three independent experiments, the error bars are given with standard deviations.

8. Specificity of MTC- and *M. smegmatis*-specific sensors with synthetic analytes

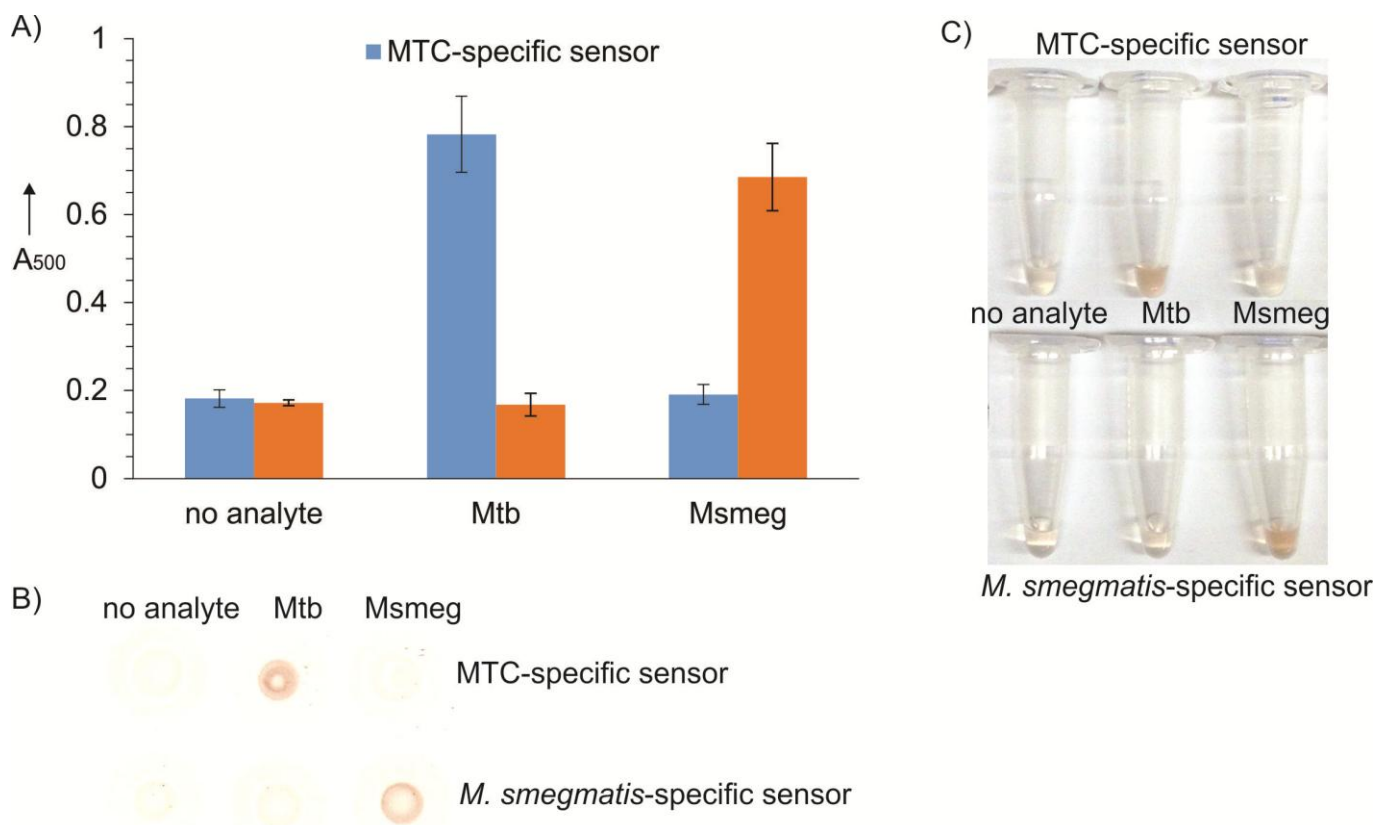


Figure S7. Specificity of MTC- and *M. smegmatis*-specific deoxyribozyme sensors with synthetic DNA analytes. A) Absorbance at 500 nm (A_{500}) for the sensors in the absence of analytes or in the presence of Mtb or Msmeg synthetic analytes (5 nM). B) TLC plate-based visualization of the samples described in panel (A). C) Visual observation of the color change for the samples described in panel (A).

9. References

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